6/5/2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025– April 2025		
Maximum Marks: 50 **Examination: End-Semester Examination** **Duration: 2 Hrs.**		
Programme code: 01 **Programme: B.Tech Computer Engineering**	**Class:**SY	**Semester:** IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Center:** COMP	
Course Code: 216U01C403	**Name of the Course: Relational Database Management Systems**	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Describe the three-schema architecture of a database system. Why are mappings between the schema levels necessary	**05**
ii)	State the informal guidelines for relation schema design. Illustrate how violations of these guidelines may be harmful.	**05**
iii)	What is a view in SQL? Explain how views are created and used. Discuss any two advantages of using views in a database system with suitable examples	**05**
Q.2	Attempt any **one**	**10**
i)	Consider the following relational Schema: member(memb no, name) book(isbn, title, authors, publisher) borrowed(memb no, isbn, date) Write SQL queries for the following: a) Find the member number and name of each member who has borrowed at least one book published by "McGraw-Hill". b) Find the member number and name of each member who has borrowed every book published by "McGraw-Hill" c) For each publisher, find the member number and name of each member who has borrowed more than five books from "McGraw-Hill" publisher	
ii)	Construct the B+ Tree for the given input: A PARTS file with Part# as the key field includes records with the following Part# values:20, 11, 14, 25, 30,12, 22, 23, 24	

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	Suppose that the search field values are inserted in the give order in a B+ tree with order of internal node m=3 and order of leaf node m_{leaf}=2. Show how the tree will expand and what the final tree will look like	
Q.3	Attempt any one	10
i)	a) Explain the basic steps of query processing with a suitable diagram. b) Draw a state diagram and discuss the typical states that a transaction goes through during execution.	
ii)	Explain the process of mapping an Entity-Relationship (ER) model to the Relational model. Illustrate with an example.	
Q.4	Attempt the following	10
	Consider the three transactions T1, T2, and T3, and the schedules S1 and S2 given below. Draw the serializability (precedence) graphs for S1 and S2, and state whether each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s). T1: r1 (X); r1 (Z); w1 (X); T2: r2 (Z); r2 (Y); w2 (Z); w2 (Y); T3: r3 (X); r3 (Y); w3 (Y); S1: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); w3 (Y);r2 (Y); w2 (Z); w2 (Y); S2: r1 (X); r2 (Z); r3 (X); r1 (Z); r2 (Y); r3 (Y); w1 (X);w2 (Z); w3 (Y); w2 (Y);	
Q.5	Attempt the following	10
	Consider the universal relation R={A,B,C,D,E,F,G,H,I,J}and the set following functional dependencies AB->C A->DE B->F F->GH D->IJ What is the key for R? Decompose R into 2NF and then 3NF relations.	